Acid–Base Titration Curves

Equivalence points, pH during a titration, and indicators

Name _____ Class _____ Date _____

INDICATORS (COLOUR-CHANGE RANGE, APPROX.)

Methyl orange: pH 3.1–4.4 Bromothymol blue: pH 6.0–7.6

Phenolphthalein: pH 8.2–10.0

Give pH to two decimal places.

A. Understanding

1. Explain the difference between the **equivalence point** and the **end point** of a titration.

2. Predict whether the pH at the equivalence point is below 7, equal to 7, or above 7 for each titration:

(a) HNO_3 with KOH (b) formic acid (HCOOH) with NaOH (c) HCl with ammonia

B. Strong acid with strong base

20.00 mL of 0.150 M HCl is titrated with 0.100 M NaOH.

3. What volume of NaOH is needed to reach the equivalence point?

4. What is the pH before any NaOH is added?

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5. What is the pH after 10.00 mL of NaOH has been added?

6. What is the pH after 40.00 mL of NaOH has been added?

C. Weak acids and indicators

7. During the titration of an unknown weak acid, the pH at the half-equivalence point is 4.20. What are the pKa and K_a of the acid?

8. Which indicator from the data box would you choose for titrating a weak acid with NaOH? Explain your choice.

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